

Introduction to Services in Angular

In Angular, services are used to encapsulate business logic, data sharing, and reusable functionality that can be shared across components. A service is a class annotated with the `@Injectable` decorator, which makes it eligible for dependency injection. This means Angular can provide instances of the service to components or other services as needed.

Common use cases for services include:

- Fetching data from APIs
- Sharing data between components
- Managing state
- Implementing utility functions

By following the dependency injection (DI) pattern, services help keep components clean and focused on their primary responsibilities, promoting code reusability and testability.

Simple Example of a Service in Angular

Let's create a simple service that provides a list of users and use it in a component.

1. Generate a Service

```
ng generate service service name ng g s user
```

2. Implement service logic : user.service.ts

```
import { Injectable } from '@angular/core';

@Injectable({
  providedIn: 'root' // Makes the service available globally
})
export class UserService {
  private users = ['Alice', 'Bob', 'Charlie'];

  constructor() { }

  // Method to get users
  getUsers(): string[] {
    return this.users;
  }
}
```

3. Component Implementation: App.component.ts

```
import { Component, OnInit } from '@angular/core';  
import { UserService } from '../user.service';
```

```
@Component({  
  selector: 'app-root',  
  template: `  
    <h1>User List</h1>  
    <ul>  
      <li *ngFor="let user of users">{{ user }}</li>  
    </ul>  
  `;  
  styleUrls: ['./app.component.css']  
})  
export class AppComponent implements OnInit {  
  users: string[] = [];  
  
  constructor(private userService: UserService) {}  
  
  ngOnInit(): void {  
    // Fetch users from the service  
    this.users = this.userService.getUsers();  
  }  
}
```

How It Works

1. The UserService contains a method getUsers to provide user data.
2. The service is injected into the AppComponent via the constructor.
3. Inside ngOnInit, the AppComponent calls the service method to get the list of users and display them in the template.